

SQL PROJECT ON HR_EMPLOYEE_ATTRITION

KPI'S

1.Total Employees

```
SELECT COUNT(*) AS total_employees FROM hr_attrition;
```

	total_employees bigint
1	1470

2.Total Employees Who Left

```
SELECT COUNT(*) AS employees_left FROM hr_attrition WHERE Attrition = 'Yes';
```

	employees_left bigint
1	237

3.Total Employees Who Stayed

```
SELECT COUNT(*) AS employees_stayed FROM hr_attrition WHERE Attrition = 'No';
```

	employees_stayed bigint
1	1233

4.Average Monthly Income

```
SELECT AVG(Monthly_Income) AS avg_monthly_income FROM hr_attrition;
```

	avg_monthly_income numeric
1	6502.9312925170068027

5.Average Age

```
SELECT AVG(Age) AS avg_age FROM hr_attrition;
```

	avg_age numeric	
1	36.9238095238095238	

6. Total Employees by Department

```
SELECT Department, COUNT(*) AS total_employees FROM hr_attrition GROUP BY Department;
```

	department character varying (30)	total_employees bigint
1	Human Resources	63
2	Research & Developm...	961
3	Sales	446

7. Employees Who Left, by Department

```
SELECT Department, COUNT(*) AS attrition_count FROM hr_attrition
WHERE Attrition = 'Yes'
GROUP BY Department ORDER BY attrition_count DESC;
```

	department character varying (30)	attrition_count bigint
1	Research & Developm...	133
2	Sales	92
3	Human Resources	12

8. Total Employees by Gender

```
SELECT Gender, COUNT(*) AS total_employees FROM hr_attrition GROUP BY Gender;
```

	gender character varying (10)	total_employees bigint
1	Female	588
2	Male	882

9. Employees Who Left, by Gender

```
SELECT Gender, COUNT(*) AS attrition_count FROM hr_attrition
WHERE Attrition = 'Yes' GROUP BY Gender;
```

	gender character varying (10)	attrition_count bigint
1	Female	87
2	Male	150

10. Employees Who Left, by JobRole

```
SELECT Job_Role, COUNT(*) AS attrition_count FROM hr_attrition
WHERE Attrition = 'Yes'
GROUP BY Job_Role ORDER BY attrition_count DESC;
```

	job_role character varying (40)	attrition_count bigint
1	Laboratory Technician	62
2	Sales Executive	57
3	Research Scientist	47
4	Sales Representative	33
5	Human Resources	12
6	Manufacturing Director	10
7	Healthcare Representati...	9
8	Manager	5
9	Research Director	2

11. Average Monthly Income by Department

```
SELECT Department, AVG(Monthly_Income) AS avg_income FROM hr_attrition
GROUP BY Department ORDER BY avg_income DESC;
```

	department character varying (30)	avg_income numeric
1	Sales	6959.1726457399103139
2	Human Resources	6654.5079365079365079
3	Research & Developm...	6281.2528616024973985

12. Employees Who Left, by Marital Status

```
SELECT Marital_Status, COUNT(*) AS attrition_count FROM hr_attrition
WHERE Attrition = 'Yes'
GROUP BY Marital_Status ORDER BY attrition_count DESC;
```

	marital_status character varying (10)	attrition_count bigint
1	Single	120
2	Married	84
3	Divorced	33

13. Top 10 Highest Paid Employees

```
SELECT Employee_Number, Job_Role, Department, Monthly_Income
FROM hr_attrition
ORDER BY Monthly_Income DESC
```

LIMIT 10;

	employee_number [PK] integer	job_role character varying (40)	department character varying (30)	monthly_income integer
1	259	Manager	Research & Developm...	19999
2	1035	Research Director	Research & Developm...	19973
3	1191	Manager	Research & Developm...	19943
4	226	Manager	Research & Developm...	19926
5	787	Manager	Research & Developm...	19859
6	1282	Manager	Sales	19847
7	1038	Manager	Sales	19845
8	1740	Manager	Sales	19833
9	1255	Research Director	Research & Developm...	19740
10	1338	Manager	Human Resources	19717

14.Average Distance From Home by Attrition

```
SELECT Attrition, ROUND(AVG(Distance_From_Home),2) AS avg_distance  
FROM hr_attrition  
GROUP BY Attrition;
```

	attrition character varying (3)	avg_distance numeric
1	No	8.92
2	Yes	10.63

15.Average Work Life Balance by department

```
SELECT department, AVG(Work_Life_Balance) AS avg_worklife_balance  
FROM hrattrition GROUP BY department order by avg_worklife_balance desc;
```

	department character varying (30)	avg_worklife_balance numeric
1	Human Resources	2.9206349206349206
2	Research & Developm...	2.7252861602497399
3	Sales	2.8161434977578475

16.Average Job Involvement by Department

```
SELECT Department, AVG(JobInvolvement) AS avg_job_involvement FROM hr_attrition  
GROUP BY Department ORDER BY avg_job_involvement DESC;
```

	department character varying (30)	avg_job_involvement numeric
1	Human Resources	2.7460317460317460
2	Research & Developm...	2.7419354838709677
3	Sales	2.7017937219730942

17. Average Years in Current Role by Job Role

```
SELECT Job_Role, AVG(Years_In_Current_Role) AS avg_years_in_role
```

```
FROM hr_attrition
```

```
GROUP BY Job_Role ORDER BY avg_years_in_role DESC;
```

	job_role character varying (40)	avg_years_in_role numeric
1	Manager	6.4509803921568627
2	Research Director	6.2875000000000000
3	Manufacturing Director	4.9724137931034483
4	Healthcare Representati...	4.8625954198473282
5	Sales Executive	4.8558282208588957
6	Research Scientist	3.2739726027397260
7	Laboratory Technician	3.2007722007722008
8	Human Resources	3.1346153846153846
9	Sales Representative	2.0120481927710843

18. Employees Who Left, by Education Field

```
SELECT Education_Field, COUNT(*) AS attrition_count FROM hr_attrition
```

```
WHERE Attrition = 'Yes'
```

```
GROUP BY Education_Field ORDER BY attrition_count DESC;
```

	education_field character varying (30)	attrition_count bigint
1	Life Sciences	89
2	Medical	63
3	Marketing	35
4	Technical Degree	32
5	Other	11
6	Human Resources	7

19. Classify Employees into Income Bands using CASE WHEN

```
SELECT
```

```

CASE WHEN Monthly_Income >= 15000 THEN 'High Income'
      WHEN Monthly_Income >= 5000 THEN 'Medium Income'
      ELSE 'Low Income' END AS income_band,
COUNT(*) AS total_employees FROM hr_attrition
GROUP BY
CASE WHEN Monthly_Income >= 15000 THEN 'High Income'
      WHEN Monthly_Income >= 5000 THEN 'Medium Income'
      ELSE 'Low Income' END;

```

	income_band text	total_employees bigint
1	High Income	133
2	Medium Inco...	588
3	Low Income	749

20. Using a CTE, find departments with more employees than the average department size

```
WITH dept_count AS
```

```
( SELECT Department, COUNT(*) AS total_employees FROM hr_attrition
```

```
GROUP BY Department )
```

```
SELECT * FROM dept_count WHERE total_employees > (SELECT AVG(total_employees)
FROM dept_count);
```

	department character varying (30)	total_employees bigint
1	Research & Developm...	961

21. Using RANK(), rank departments by average monthly income

```
SELECT Department,
```

```
AVG(Monthly_Income) AS avg_income,
```

```
RANK() OVER (ORDER BY AVG(Monthly_Income) DESC) AS rnk
```

FROM hr_attrition GROUP BY Department;

	department character varying (30)	avg_income numeric	mk bigint
1	Sales	6959.1726457399103139	1
2	Human Resources	6654.5079365079365079	2
3	Research & Developm...	6281.2528616024973985	3

22. For each job role, total employees and percentage contribution to overall headcount

SELECT Job_Role, COUNT() AS total_employees,

ROUND(COUNT() * 100.0 / (SELECT COUNT(*) FROM hr_attrition), 2) AS pct_contribution

FROM hr_attrition

GROUP BY Job_Role ORDER BY total_employees DESC;

	job_role character varying (40)	total_employees bigint	pct_contribution numeric
1	Sales Executive	326	22.18
2	Research Scientist	292	19.86
3	Laboratory Technician	259	17.62
4	Manufacturing Director	145	9.86
5	Healthcare Representati...	131	8.91
6	Manager	102	6.94
7	Sales Representative	83	5.65
8	Research Director	80	5.44
9	Human Resources	52	3.54

23. Using a CTE, find job roles where average Years Since Last Promotion is above the company average

WITH role_promotion AS (

SELECT Job_Role, AVG(Years_Since_Last_Promotion) AS avg_promo_gap

FROM hr_attrition GROUP BY Job_Role)

SELECT * FROM role_promotion WHERE avg_promo_gap > (SELECT AVG(avg_promo_gap)

FROM role_promotion);

	job_role character varying (40)	avg_promo_gap numeric
1	Manager	4.8333333333333333
2	Healthcare Representati...	2.9694656488549618
3	Sales Executive	2.4846625766871166
4	Research Director	3.1875000000000000

24. Using a CTE and RANK(), rank job roles within each department by average income

WITH role_income AS (

SELECT Department, Job_Role, ROUND(AVG(Monthly_Income),2) AS avg_income,

RANK() OVER (PARTITION BY Department ORDER BY AVG(Monthly_Income) DESC) AS rnk

FROM hr_attrition GROUP BY Department, Job_Role)

SELECT * FROM role_income WHERE rnk <= 3;

	department character varying (30)	job_role character varying (40)	avg_income numeric	rnk bigint
1	Human Resources	Manager	18088.64	1
2	Human Resources	Human Resources	4235.75	2
3	Research & Developm...	Manager	17130.33	1
4	Research & Developm...	Research Director	16033.55	2
5	Research & Developm...	Healthcare Representati...	7528.76	3
6	Sales	Manager	16986.97	1
7	Sales	Sales Executive	6924.28	2
8	Sales	Sales Representative	2626.00	3